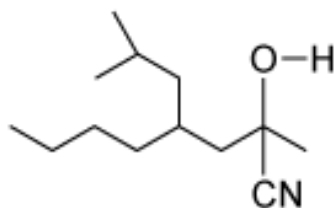


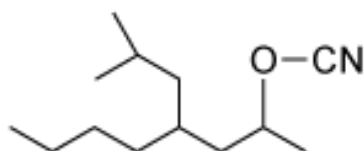
Which structure is the product of the reaction of 4-*tert*-butyl-2-octanone and NaCN under acidic conditions?

☐ A.



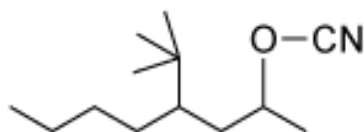
[10%]

☐ B.



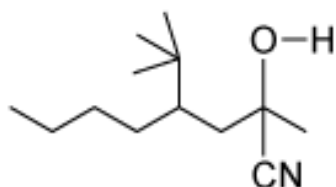
[6%]

☐ C.



[18%]

☒ D.

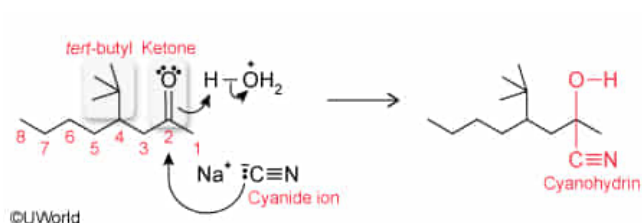


[63%]

Correct

64% answered correctly

Explanation:



Molecular structure determines how a reaction will proceed. The following steps describe how to determine the structure of an organic molecule from its name:

1. Determine the number of carbon atoms in the **parent chain**, which is the

longest chain containing the **highest priority functional group**. The **prefix** of the parent alkane determines the number of carbons, and the **suffix** describes the high priority functional group. The number prior to the chain name describes the carbon to which the main functional group is attached.

2. Determine the **position and identity** of each **substituent**, if any. All **substituent names** are placed at the **beginning** of the **compound name**, together with the number of the carbon to which the substituent is attached.

The reactant 4-*tert*-butyl-2-octanone has an eight-carbon parent chain denoted by the prefix *oct*–, and the suffix –*one* indicates the compound is a ketone with the carbonyl at carbon 2. The substituent is a *tert*-butyl on carbon 4.

The **reaction of a ketone** or an **aldehyde** with a **cyanide ion** (from NaCN) acting as a nucleophile and attacking the electrophilic carbonyl (C=O) carbon produces a **cyanohydrin**. The cyanohydrin functional group contains a hydroxyl group (–OH) and a cyano group (C≡N) attached to the same carbon.

(Choice A) This structure shows the correct addition of CN[–] to the carbonyl; however, the compound contains an isobutyl group instead of a *tert*-butyl on C4.

(Choices B and C) These structures show the addition of CN[–] to the carbonyl oxygen instead of the carbonyl carbon. The starting material from Choice B contains an isobutyl group instead of a *tert*-butyl on C4.

Educational objective:

The reaction of a ketone or aldehyde with sodium cyanide (NaCN) results in the nucleophilic attack of the carbonyl by the cyanide ion, and produces a cyanohydrin. Organic molecules are named after the number of carbon atoms in the longest carbon chain, adding the suffix of the highest priority functional group. Substituent names precede the chain name, and the location of the high-priority group is denoted by placing its location between the substituent and chain names.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401252

System:
Functional Groups and Their Reactions

Ethanol is an important source of energy that is frequently used as a supplement to gasoline. It can be obtained from renewable resources such as the cellulose found in plant matter. When subjected to a process called gasification, the plant mass is converted to a mixture called syngas, composed predominantly of carbon monoxide and molecular hydrogen. Syngas can be passed over a rhodium catalyst to generate ethanol and other products. After using this method, a group of researchers analyzed the resulting products by gas chromatography using helium as a carrier gas. The results are shown in Figure 1.

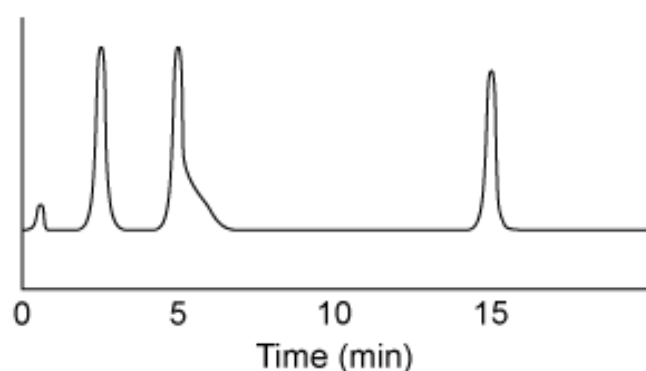
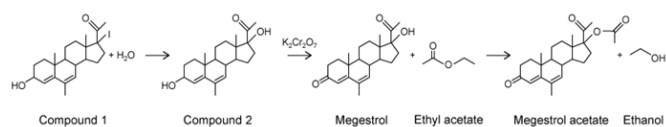


Figure 1 Gas chromatograph of syngas catalysis products

Mass spectrometry revealed that the process yielded similar amounts of ethanol, acetic acid, and acetaldehyde, with trace amounts of ethane. In addition, the ethanol peak was found to be contaminated with ethyl acetate. Another group of researchers in the same laboratory had recently produced Compound 1 as a by-product in a separate reaction. They combined Compound 1 with purified ethyl acetate to form megestrol acetate, a drug used in the treatment of certain cancers. This reaction also produced ethanol, as shown in Scheme 1.



Scheme 1

In Scheme 1, conversion of Compound 1 to Compound 2 most likely begins with which of the following steps?

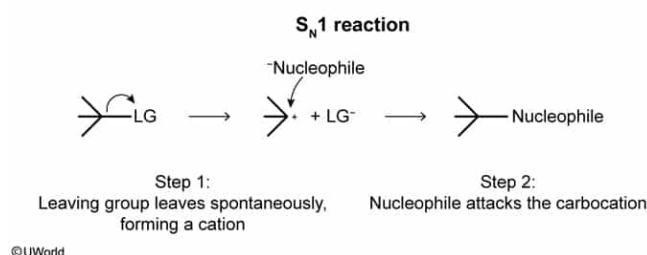
- ☐ A. Nucleophilic attack by water, causing iodide to leave [28%]

- ☐ B. Deprotonation of water to enhance its nucleophilicity [12%]
- ✓ ☒ C. Formation of a carbocation when iodide leaves [52%]
- ☐ D. Protonation of iodide to form a better leaving group [5%]

Correct

53% answered correctly

Explanation:



Nucleophilic substitution reactions can occur through one of **two mechanisms**: S_N1 or S_N2. **S_N1 reactions** proceed in **two steps** and involve the formation of a **carbocation intermediate**. In these reactions, the leaving group leaves *before* the nucleophile attacks. **S_N2 reactions** occur in a single step in which the nucleophile attacks and the leaving group leaves simultaneously.

Tertiary leaving groups are more likely to **undergo S_N1 reactions** than S_N2 reactions for two reasons. First, they are **sterically hindered** by the bulky alkyl groups surrounding them, and these groups impede nucleophilic attack. Second, tertiary carbon atoms can form relatively **stable carbocations** compared with secondary or primary carbon atoms. The positive charge associated with carbocations enhances their electrophilicity, helping them overcome the steric hindrance that otherwise inhibits nucleophilic attack.

The conversion of Compound 1 to Compound 2 entails the removal of an iodide ion from a tertiary carbon and its replacement by a hydroxyl group. Because a tertiary carbon is involved in the reaction, it will most likely proceed by an S_N1 mechanism. Therefore, the first step will be the formation of a carbocation when iodide leaves.

(Choice A) Nucleophilic attack is the first step in an S_N2 reaction but not in an S_N1 reaction.

(Choice B) Deprotonation of water would enhance its nucleophilicity, but the first step of an S_N1 reaction is always the formation of a carbocation as the bond to a leaving group is broken.

(Choice D) Halides such as iodide generally cannot form more than one bond at a time. Therefore, iodide would be unable to be protonated before it leaves the tertiary carbon.

Educational objective:

Nucleophilic substitution reactions can proceed by S_N1 or S_N2 mechanisms.

Tertiary carbons are more likely to participate in S_N1 reactions due to steric

hindrance and the ability to form stable carbocations.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401253

System:
Functional Groups and Their Reactions

Ethanol is an important source of energy that is frequently used as a supplement to gasoline. It can be obtained from renewable resources such as the cellulose found in plant matter. When subjected to a process called gasification, the plant mass is converted to a mixture called syngas, composed predominantly of carbon monoxide and molecular hydrogen. Syngas can be passed over a rhodium catalyst to generate ethanol and other products. After using this method, a group of researchers analyzed the resulting products by gas chromatography using helium as a carrier gas. The results are shown in Figure 1.

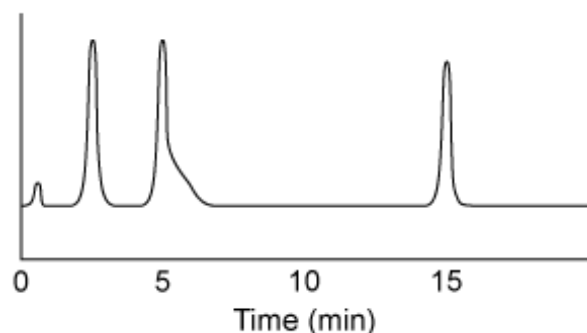
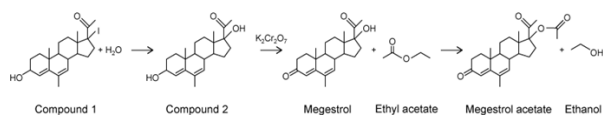


Figure 1 Gas chromatograph of syngas catalysis products

Mass spectrometry revealed that the process yielded similar amounts of ethanol, acetic acid, and acetaldehyde, with trace amounts of ethane. In addition, the ethanol peak was found to be contaminated with ethyl acetate. Another group of researchers in the same laboratory had recently produced Compound 1 as a by-product in a separate reaction. They combined Compound 1 with purified ethyl acetate to form megestrol acetate, a drug used in the treatment of certain cancers. This reaction also produced ethanol, as shown in Scheme 1.

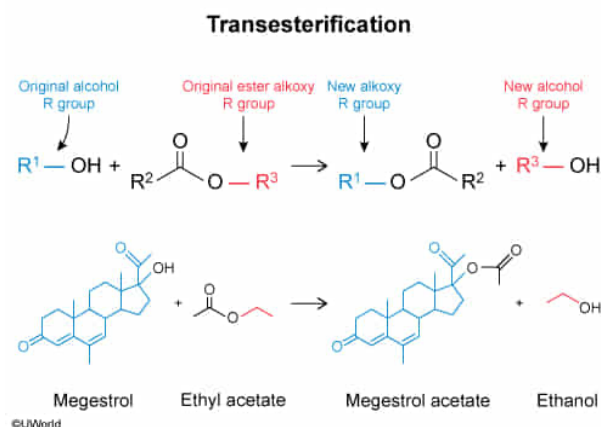


Scheme 1

The conversion of megestrol to megestrol acetate shown in Scheme 1 represents which kind of reaction?

- ✔ ☒ A. Transesterification [77%]
- ☐ B. Saponification [4%]
- ☐ C. Aldol condensation [10%]
- ☐ D. Oxidation [7%]

Explanation:



Transesterification is the reaction between an **ester and an alcohol** that results in an **exchange of alkoxy and alcohol R groups**. The reaction can be either **acid-catalyzed** or **base-catalyzed**, and involves nucleophilic attack by the alcohol on the carbonyl carbon of the ester to form a new alkoxy group. The original alkoxy group leaves and becomes protonated to form a new alcohol.

The conversion of megestrol to megestrol acetate described in Scheme 1 is a reaction between an alcohol (megestrol) and an ester (ethyl acetate). It results in the formation of a bond between megestrol and acetate, and the ethoxy group leaves as ethanol. This is an example of transesterification.

(Choice B) **Saponification** is the removal of an alkoxy group from an ester by a base. It forms a carboxylic acid and an alcohol rather than a new ester and an alcohol.

(Choice C) An **aldol condensation** is a carbon-carbon bond forming reaction that requires two carbonyl compounds (ketones and/or aldehydes) rather than an ester and an alcohol. It forms an α,β -unsaturated carbonyl compound rather than an ester and an alcohol.

(Choice D) Oxidation is the loss of electrons by an atom. In organic molecules, it generally results in fewer C-H bonds and more C-O bonds, neither of which occurs in the described process.

Educational objective:

Transesterification is the exchange of alcohol and alkoxy R groups between an ester and an alcohol.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 401254

System:

Functional Groups and Their Reactions

Ethanol is an important source of energy that is frequently used as a supplement to gasoline. It can be obtained from renewable resources such as the cellulose found in plant matter. When subjected to a process called gasification, the plant mass is converted to a mixture called syngas, composed predominantly of carbon monoxide and molecular hydrogen. Syngas can be passed over a rhodium catalyst to generate ethanol and other products. After using this method, a group of researchers analyzed the resulting products by gas chromatography using helium as a carrier gas. The results are shown in Figure 1.

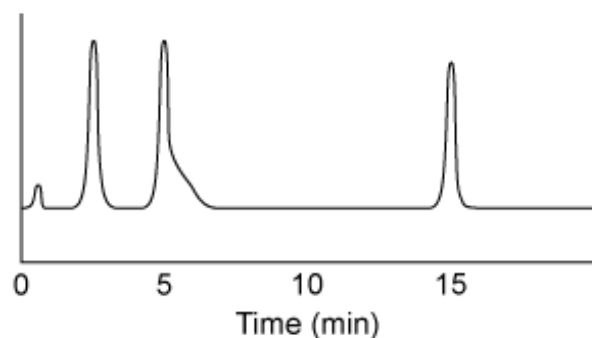
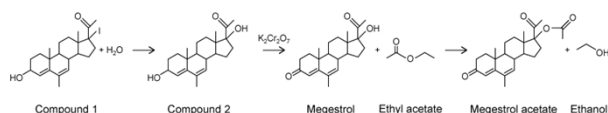


Figure 1 Gas chromatograph of syngas catalysis products

Mass spectrometry revealed that the process yielded similar amounts of ethanol, acetic acid, and acetaldehyde, with trace amounts of ethane. In addition, the ethanol peak was found to be contaminated with ethyl acetate. Another group of researchers in the same laboratory had recently produced Compound 1 as a by-product in a separate reaction. They combined Compound 1 with purified ethyl acetate to form megestrol acetate, a drug used in the treatment of certain cancers. This reaction also produced ethanol, as shown in Scheme 1.



Scheme 1

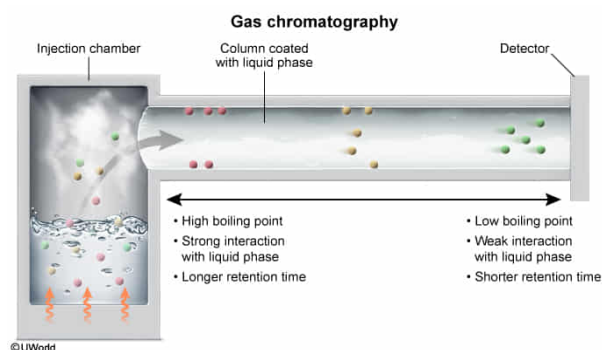
Acetic acid most likely eluted from the gas chromatograph at what time point?

- ☐ A. 1 min [6%]
- ☐ B. 2 min [12%]
- ☐ C. 5 min [18%]
- ☒ D. 15 min [63%]

Correct

62% answered correctly

Explanation:



Gas chromatography (GC) separates molecules primarily on **the basis of boiling point**. The injection chamber is heated to a temperature above the boiling point of all compounds being analyzed (usually 200°C–300°C), causing all components to vaporize. The walls of the GC column are coated with a thin layer of liquid that acts as the stationary phase. An inert gas flows through the column, carrying other gases with it. Compounds with a **lower boiling point** tend to stay in the gas phase and **move through the column quickly**, whereas compounds with **higher boiling points** have a greater tendency to interact with the liquid phase and a **longer retention time**.

The chromatogram shown in Figure 1 has four peaks. To determine which one corresponds to acetic acid, the boiling point of acetic acid relative to the other compounds must be determined. Boiling point is a function of intermolecular forces, with fewer forces corresponding to lower boiling points. Organic molecules with the same number of carbon atoms follow a **pattern of increasing intermolecular forces** and boiling points: **Alkanes** have the weakest intermolecular forces and the **lowest boiling points**, followed by **aldehydes and ketones**, then **alcohols**, and **carboxylic acids**, which have the **highest boiling points**. Acetic acid is a carboxylic acid and has the highest boiling point of the compounds produced; therefore, it has the longest retention time and must elute at 15 min.

(Choice A) The first peak (1 min) corresponds to ethane, which has the lowest boiling point.

(Choice B) The second peak (2 min) corresponds to acetaldehyde.

(Choice C) The third peak (5 min) corresponds to ethanol. The contaminating ethyl acetate, which has a similar boiling point to ethanol, is also in this peak.

Educational objective:

Gas chromatography separates compounds primarily by boiling point. Molecules with low boiling points elute before those with high boiling points. For compounds with the same number of carbon atoms, alkanes have the lowest boiling point, followed by aldehydes and ketones, alcohols, and carboxylic acids.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401255

System:
Functional Groups and Their Reactions

Ethanol is an important source of energy that is frequently used as a supplement to gasoline. It can be obtained from renewable resources such as the cellulose found in plant matter. When subjected to a process called gasification, the plant mass is converted to a mixture called syngas, composed predominantly of carbon monoxide and molecular hydrogen. Syngas can be passed over a rhodium catalyst to generate ethanol and other products. After using this method, a group of researchers analyzed the resulting products by gas chromatography using helium as a carrier gas. The results are shown in Figure 1.

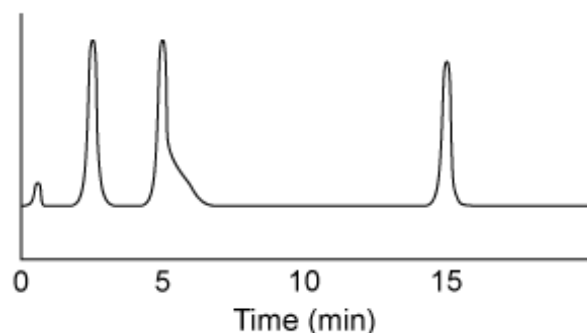
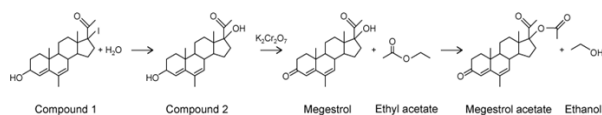


Figure 1 Gas chromatograph of syngas catalysis products

Mass spectrometry revealed that the process yielded similar amounts of ethanol, acetic acid, and acetaldehyde, with trace amounts of ethane. In addition, the ethanol peak was found to be contaminated with ethyl acetate. Another group of researchers in the same laboratory had recently produced Compound 1 as a by-product in a separate reaction. They combined Compound 1 with purified ethyl acetate to form megestrol acetate, a drug used in the treatment of certain cancers. This reaction also produced ethanol, as shown in Scheme 1.



Scheme 1

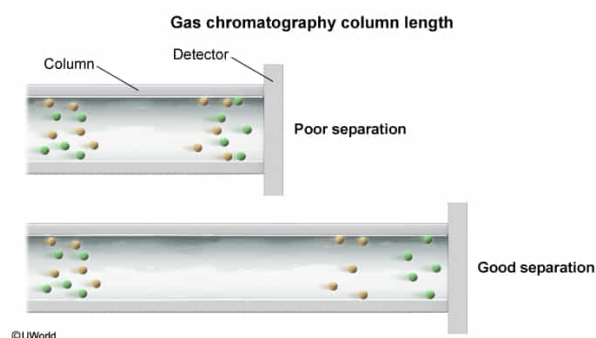
Ethyl acetate could be better separated from ethanol by doing which of the following?

- ☐ A. Replacing helium with nitrogen as the carrier gas [7%]
- ☒ B. Running the mixture through a longer column [68%]
- ☐ C. Starting the chromatograph at a higher temperature [20%]
- ☐ D. Increasing the carrier gas flow rate [3%]

Correct

69% answered correctly

Explanation:



Chromatography methods **separate molecules** based on their relative **affinities for a stationary phase** versus a mobile phase. For compounds to separate efficiently on any column, they need sufficient **time to interact** with the stationary phase. Given enough time, even subtle differences in affinity for the stationary phase can be amplified, allowing separation of compounds with similar properties. Increasing the length of the stationary phase through which the compounds must travel can provide the necessary time.

Ethanol and ethyl acetate have similar, but not identical, boiling points and can be separated on a gas chromatograph by running them through a longer column.

(Choice A) Helium and nitrogen are both inert gases with low boiling points, making them good mobile phases for gas chromatography because they do not interact with the stationary phase or the compounds being analyzed. Replacing one with the other is unlikely to alter interactions with the stationary phase and is unlikely to improve **resolution**.

(Choices C and D) Causing the carrier gas to move through the column more quickly reduces the time available for compounds to interact with the stationary phase. Increasing the temperature of the gas increases the flow rate, which leads to a lower resolution of compounds.

Educational objective:

Chromatography techniques separate compounds based on their affinity for a stationary phase over a mobile phase. In general, adjusting experimental parameters such that molecules have more time to interact with the stationary phase improves the resolution of each compound.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 401256

System:

Functional Groups and Their Reactions

Ethanol is an important source of energy that is frequently used as a supplement to gasoline. It can be obtained from renewable resources such as the cellulose found in plant matter. When subjected to a process called gasification, the plant mass is converted to a mixture called syngas, composed predominantly of carbon monoxide and molecular hydrogen. Syngas can be passed over a rhodium catalyst to generate ethanol and other products. After using this method, a group of researchers analyzed the resulting products by gas chromatography using helium as a carrier gas. The results are shown in Figure 1.

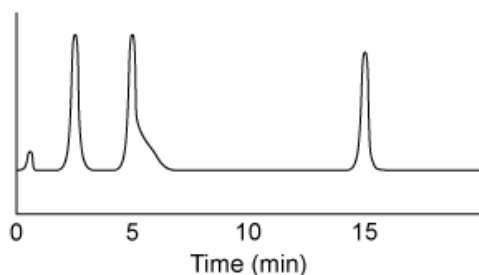
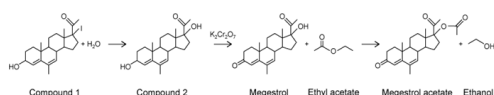
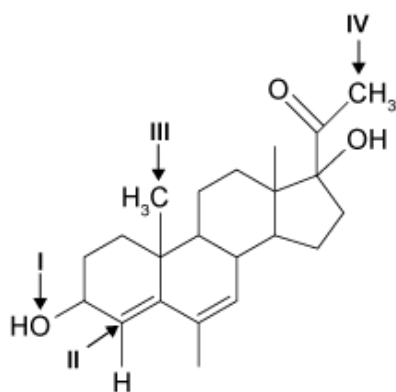


Figure 1 Gas chromatograph of syngas catalysis products

Mass spectrometry revealed that the process yielded similar amounts of ethanol, acetic acid, and acetaldehyde, with trace amounts of ethane. In addition, the ethanol peak was found to be contaminated with ethyl acetate. Another group of researchers in the same laboratory had recently produced Compound 1 as a by-product in a separate reaction. They combined Compound 1 with purified ethyl acetate to form megestrol acetate, a drug used in the treatment of certain cancers. This reaction also produced ethanol, as shown in Scheme 1.



Scheme 1



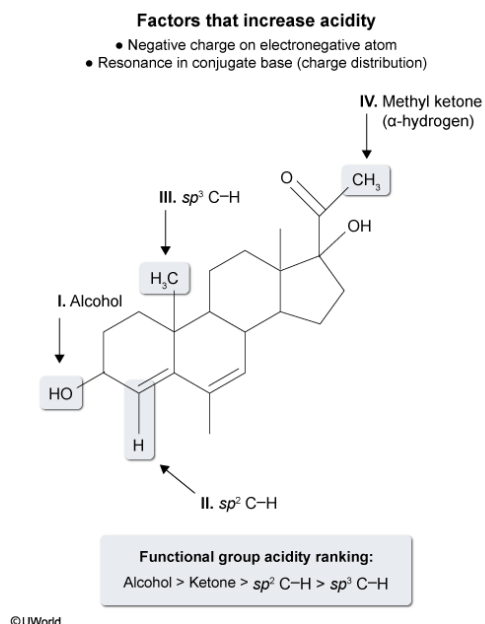
What atom indicated would be most readily deprotonated if Compound 2 reacts with NaOH?

- ✓ ☒ A. I [52%]
☐ B. II [12%]
☐ C. III [2%]
☐ D. IV [32%]

Correct

52% answered correctly

Explanation:



For a compound in solution, as the pH is increased, the most acidic **functional group** of the compound will be deprotonated first. The **acidity** of a functional group is determined largely by the **electronegativity** of the proton-donating atom and by the ability to **stabilize the negative charge** in the corresponding **conjugate base** through charge distribution (atom size) or delocalization (**resonance**). Negative charge is stabilized better on more electronegative atoms, and negative charge that can participate in resonance will be stabilized further by being delocalized across more than one atom. Sites that better stabilize a negative charge are more likely to be deprotonated (more acidic).

Atoms I–IV in Compound 2 are oxygen (alcohol), sp^2 carbon (alkene), sp^3 carbon (alkane), and methyl ketone (α -H), respectively. Upon deprotonation, the negatively charged atom in the conjugate base of these functional groups is oxygen for alcohols and carbon for the others. Because oxygen is more electronegative than carbon, a negative charge on oxygen in the conjugate base of the alcohol is more stabilized than the charge on the carbon atoms in the three other functional groups. Therefore, the alcohol oxygen atom has the most acidic proton and will be deprotonated when reacted with NaOH.

(Choices B and C) Alkene protons (sp^2 C–H) are more acidic than alkane protons (sp^3 C–H), but both are less acidic than an alcohol proton because the conjugate bases of alkenes and alkanes are less stable, with the negative charge concentrated on a less electronegative carbon atom.

(Choice D) Ketones are less acidic than alcohols but more acidic than hydrocarbons. The negative charge in the conjugate base of a ketone is delocalized by resonance across the α -carbon and carbonyl oxygen atoms, which helps stabilize the charge.

Educational objective:

Acidity of a functional group is determined by the stability of the conjugate base it forms. Factors such as the size and electronegativity of the proton-donating atom as well as the ability to stabilize the negatively charged atom in the conjugate base by charge distribution or resonance contribute to acidity.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401257

System:
Functional Groups and Their Reactions

Ethanol is an important source of energy that is frequently used as a supplement to gasoline. It can be obtained from renewable resources such as the cellulose found in plant matter. When subjected to a process called gasification, the plant mass is converted to a mixture called syngas, composed predominantly of carbon monoxide and molecular hydrogen. Syngas can be passed over a rhodium catalyst to generate ethanol and other products. After using this method, a group of researchers analyzed the resulting products by gas chromatography using helium as a carrier gas. The results are shown in Figure 1.

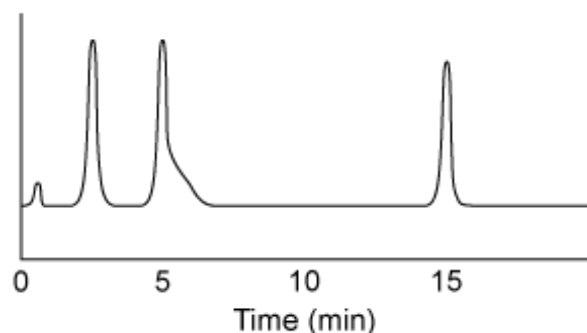
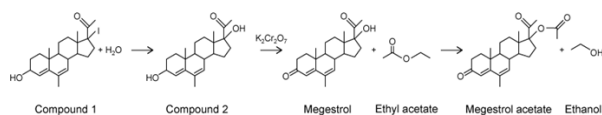


Figure 1 Gas chromatograph of syngas catalysis products

Mass spectrometry revealed that the process yielded similar amounts of ethanol, acetic acid, and acetaldehyde, with trace amounts of ethane. In addition, the ethanol peak was found to be contaminated with ethyl acetate. Another group of researchers in the same laboratory had recently produced Compound 1 as a by-product in a separate reaction. They combined Compound 1 with purified ethyl acetate to form megestrol acetate, a drug used in the treatment of certain cancers. This reaction also produced ethanol, as shown in Scheme 1.

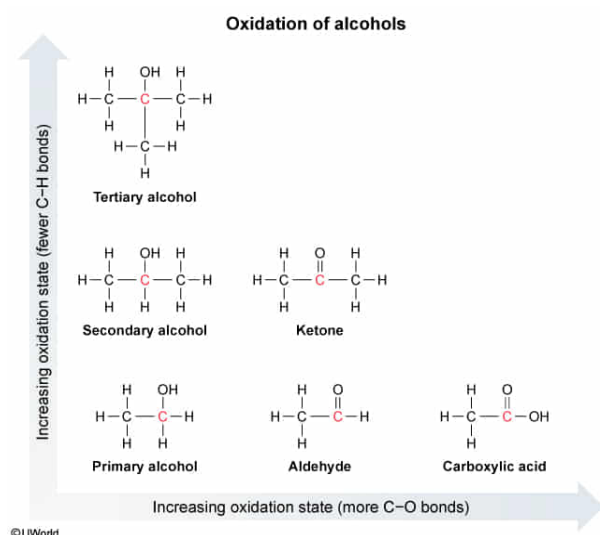


Scheme 1

During the conversion of Compound 2 to megestrol, why was the secondary alcohol affected, whereas the tertiary alcohol was not?

- ✔ ☒ A. Tertiary alcohols cannot be readily oxidized. [84%]
☐ B. The secondary alcohol is stabilized by resonance. [6%]
☐ C. The tertiary alcohol is less acidic than the secondary alcohol. [5%]
☐ D. Secondary alcohols are better nucleophiles. [3%]

Explanation:



The conversion of Compound 2 to megestrol is an **oxidation reaction**, with $K_2Cr_2O_7$ acting as the oxidizing agent. When **carbon atoms are oxidized**, they **lose bonds to hydrogen** and/or **gain bonds to oxygen**. Therefore, **oxidizing agents** convert primary alcohols to aldehydes and carboxylic acids, and secondary alcohols to ketones. However, **tertiary carbon atoms have no bonds to hydrogen**. Because no bonds to hydrogen can be lost, additional bonds to oxygen would result in a carbon atom with five bonds. Carbon can participate in only four bonds, so this is not possible. Therefore, tertiary alcohols **cannot be oxidized** by losing bonds to hydrogen *or* by gaining bonds to oxygen.

(Choice B) The double bonds in Compound 2 have **resonance forms**, but the alcohol carbon has no lone electrons or pi bonds, which are required to participate in resonance structures. In addition, resonance does not significantly contribute to a molecule's ability to be oxidized.

(Choice C) Tertiary alcohols are less acidic than secondary alcohols because tertiary alcohols are more sterically hindered, which makes its **conjugate base** less stable than that of a secondary alcohol. However, acidity does not affect a molecule's ability to be oxidized.

(Choice D) Although secondary alcohols are better nucleophiles than tertiary alcohols, a compound's nucleophilicity is not related to its ability to be oxidized.

Educational objective:

Oxidation of carbon atoms involves the loss of bonds to hydrogen and is frequently accompanied by the formation of bonds to oxygen. Primary and secondary alcohols can be oxidized to carboxylic acids and ketones, respectively, but tertiary alcohols are in their highest possible oxidation state.

Time spent:0 sec
Subject:
Organic Chemistry

QID:401258
System:
Functional Groups and Their Reactions